

Vision Transformers for Chest X-ray Disease Classification

(2023)

Problem

Multi-label classification of 14 thoracic diseases from chest radiographs.

Method

A Vision Transformer (ViT) architecture using patch embedding and self-attention, pretrained on ImageNet then fine-tuned.

Dataset

ChestX-ray14 dataset.

Evaluation

AUROC computed per disease class.

Results

The ViT model achieved a mean AUROC of 0.812 across 14 classes, modestly beating a ResNet-50 baseline at 0.795.

Limitations

Interpretability of attention maps for radiologist trust remains unresolved; clinicians in a pilot study found the maps difficult to act on.